

Orientation – Graph Mining

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네트워크 과학연구실
NETWORK SCIENCE LAB



가톨릭대학교
THE CATHOLIC UNIVERSITY OF KOREA

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Goal

- This lecture provides understandings of concepts and methodologies of graph mining
- Students will learn fundamental concepts and properties of the networked data
- Based on the concepts, the students will learn metrics and analysis methods for structural properties of graphs
- The students will obtain capabilities for analyzing unstructured big data with graphs

Contents

- Fundamental concepts of network science
- Power laws and scale-free networks
- Structure, nodes, and links analysis
- Network communities
- Node classification and link prediction
- Diffusion of information
- Influence propagation

- Students do not need to buy textbooks
- Most of the lecture materials will be provided in digital format (PDF files)

Textbooks

- William L. Hamilton, “Graph Representation Learning Book,” Morgan and Claypool, 2020
- David Easley and Jon Kleinberg, “Networks, Crowds, and Markets: Reasoning About a Highly Connected World,” Cambridge University Press, 2010
- Albert-László Barabási, “Network science,” Cambridge University Press, 2017

- **Week 1: Introduction to Graph Mining**

- Overview of graph mining
- Types of graphs and their applications
- Basic graph concepts (nodes, edges, degree, etc.)

- **Week 2: Graph Representation and Storage**

- Adjacency matrix and list
- Sparse matrix representations
- Graph databases and storage systems

- **Week 3: Centrality Measures**

- Degree centrality
- Betweenness centrality
- Closeness centrality
- Eigenvector centrality
- PageRank and its application in web search

- **Week 4: Graph Visualization**

- Visualization techniques (spring-embedded, circular, etc.)
- Tools for graph exploration and visualization (Gephi, Cytoscape, etc.)

- **Week 5: Community Detection**

- Definition of communities and their properties
- Clustering techniques (k-means, hierarchical, etc.)
- Modularity and its variants
- Tools for community detection (Louvain, Infomap, etc.)

- **Week 6: Link Prediction**

- Types of links (positive, negative, neutral)
- Commonly used link prediction methods (common neighbors, Jaccard coefficient, etc.)
- Tools for link prediction

- **Week 7: Subgraph Mining**
 - Frequent subgraph mining (FSM)
 - FSM algorithms (gSpan, FSG, etc.)
 - Tools for frequent subgraph mining (Traceminer, etc.)
- **Week 8: Mid-term Exam**
- **Week 9: Graph Kernels**
 - Overview of graph kernels
 - WL relabeling process
 - Applications of graph kernels
- **Week 10: Node Classification**
 - Overview of node classification
 - Feature extraction methods
 - Classification algorithms (SVM, Random Forest, etc.)
 - Evaluation metrics

- **Week 11: Graph Applications in Social Networks**
 - Overview of graph applications in social networks
 - Social network analysis
 - Community detection and link prediction in social networks
- **Week 12: Graph Applications in Bioinformatics**
 - Overview of graph applications in biological networks
 - Graph pattern mining
 - Graph classification task in Bioinformatics
- **Week 13: Graph Applications in Web Graph**
 - Overview of graph applications in web graph
 - Web graph crawling
 - Web graph analysis

- **Week 14: Hypergraph**
 - What is hypergraph?
 - Incidence matrix
 - Message passing on hypergraph
 - Hypergraph application
- **Week 15: Additional Topics and Wrap-up**
 - Graph neural networks
 - Graph Transformers
 - Summary of key concepts
 - Future directions in graph
- **Week 16: Final Exam**

Assessment

- Midterm exam: 30%
- Final exam: 30%
- Homework: 30%
- Attendance: 10%
- All test questions are made up within lecture slides and assignments

Tentative Grading Scale

- A: 80~100
- B: 60~80
- C: ~60
- F: Absence from exams

Weekly Programming Practice

- Practice programming in one class every week on Thursday

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